

Types of Variables and Samples

Chapter 1

Sections 1.3 and 1.5

MAT 1272

Introduction to Statistics

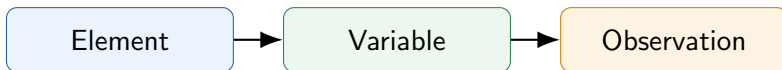
Based on the course textbook

Prem S. Mann

Introductory Statistics, 10th Edition

John Wiley & Sons

In the previous lecture, we learned the basic language of statistics.



Today we ask two important questions:

- ❶ **What kind of variable are we studying?**
- ❷ **Who are we studying?**

These two questions will appear throughout the entire course.

Not all variables behave in the same way.

The first major distinction is:

Quantitative

Qualitative

The key question

Can the variable be measured numerically in a meaningful way?

Definition

A **qualitative variable** describes a characteristic that can be classified into categories.

Examples:

- major of a college student
- type of transportation to campus
- marital status
- political party preference
- type of residence

Think: categories

The values describe **which category** an element belongs to.

Suppose we ask students:

“How did you get to campus today?”

Possible answers:

Subway

Bus

Car

Walk

Other

What type of variable?

Qualitative

Definition

A **quantitative variable** is a variable that can be measured numerically.

Examples:

- age
- income
- number of courses taken
- commute time
- height
- number of children in a household

Think: numbers

The numerical values have a meaningful mathematical interpretation.

Suppose we ask:

“How many minutes did it take you to get to campus?”

Possible observations:

Student	A	B	C	D	E
Minutes	35	52	18	70	42

What type of variable?

Quantitative

Quantitative variables can be divided into two types.

Discrete

Continuous

The question becomes:

Can we count the possible values, or can values vary continuously?

Definition

A **discrete variable** can take a countable set of values.

Discrete variables often arise from **counting**.

Examples:

- number of students in a class
- number of cars in a household
- number of courses taken
- number of calls received
- number of defective items

A useful clue

If you are counting “how many,” the variable is often discrete.

Suppose we record the number of courses taken by five students:

Student A	Student B	Student C	Student D	Student E
4	5	3	6	4

Can a student take:

4.37 courses?

Suppose we record the number of courses taken by five students:

Student A	Student B	Student C	Student D	Student E
4	5	3	6	4

Can a student take:

4.37 courses?

No

The number of courses is a **discrete** variable.

Definition

A **continuous variable** can assume any value within an interval.

Continuous variables often arise from **measurement**.

Examples:

- height
- weight
- temperature
- time
- distance

A useful clue

If you are measuring “how much” or “how long,” the variable is often continuous.

Suppose a student's commute takes:

37.5 minutes.

Could it be:

37.52 minutes?

37.523 minutes?

37.5237 minutes?

In principle, yes.

Therefore

Commute time is a **continuous** quantitative variable.

Discrete

Any value in an interval

Usually comes from counting

Number of students

Number of cars

Number of calls

Continuous

Countable values

Usually comes from measuring

Student height

Car speed

Call duration

Important

Not every variable with numbers is quantitative in the same way. We need to understand what the numbers represent.

Suppose a data set contains:

07666

07024

10001

11201

These look like numbers.

But are they a quantitative variable?

Suppose a data set contains:

07666

07024

10001

11201

These look like numbers.

But are they a quantitative variable?

No

ZIP codes are identifiers or categories.

The arithmetic operation $07666 + 07024$ does not give us a meaningful statistical quantity.

Lesson

The appearance of a number does not automatically make a variable quantitative.

Classify each variable as:

Qualitative

Quantitative

- ① Number of siblings
- ② Favorite type of music
- ③ Monthly rent
- ④ Number of text messages sent today
- ⑤ Type of housing

Bonus

For the quantitative variables, decide whether they are **discrete** or **continuous**.

Variable	Type	If quantitative
Number of siblings	Quantitative	Discrete
Favorite music	Qualitative	–
Monthly rent	Quantitative	Continuous
Number of text messages	Quantitative	Discrete
Type of housing	Qualitative	–

Remember

Qualitative → categories

Quantitative → numerical measurement

Discrete → countable

Continuous → measurable on an interval

Now we change the question.

Instead of asking:

“What kind of variable is this?”

we ask:

“Who are we studying?”

This leads to two fundamental ideas:

Population and **Sample**

Definition

A **population** is the collection of all elements whose characteristics are being studied.

Examples:

- all students at City Tech
- all households in New York City
- all registered voters in New Jersey
- all cars produced by a manufacturer
- all patients receiving a particular treatment

Definition

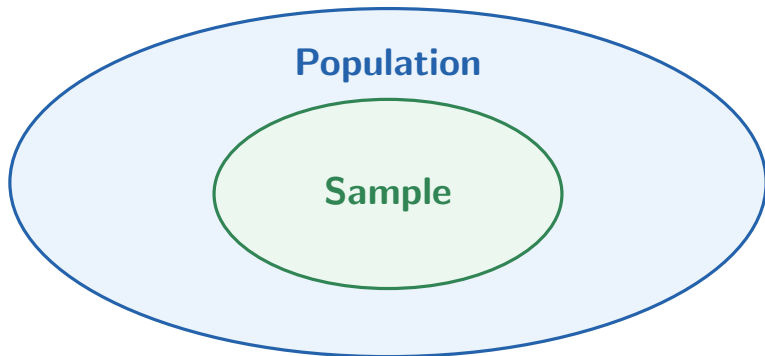
A **sample** is a portion of the population selected for study.

Why use a sample?

- The population may be too large.
- Studying everyone may be too expensive.
- Studying everyone may take too much time.
- Sometimes measurement destroys the item being studied.

The goal

Use information from a sample to learn about the population.



$$\text{Sample} \subseteq \text{Population}$$

Sometimes the population needs to be defined carefully.

Suppose we ask:

“What do college students think about online classes?”

What is the population?

Sometimes the population needs to be defined carefully.

Suppose we ask:

“What do college students think about online classes?”

What is the population?

- All students in the United States?
- All college students in New York?
- All City Tech students?
- All City Tech students currently taking online classes?

Important

Before collecting data, we need to clearly define **who belongs to the population**.

A sample should ideally reflect important characteristics of the population.



Representative sample

A sample that represents the characteristics of the population as closely as possible.

Imagine a college has:

- 60% commuters
- 40% students who live near campus

Now suppose we ask only students who live near campus:

“How long does it take you to get to campus?”

The sample may systematically underestimate commute time.

Problem

A sample can be large and still be **unrepresentative**.

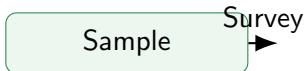
There are two different ways to collect information.

Census

Collect information from **every member of the population.**

Sample survey

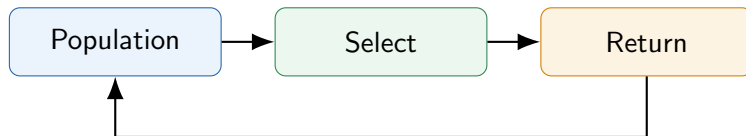
Collect information from **members of a sample.**



Suppose we select one element from a population.

Sampling with replacement

After selecting an element, we return it to the population before making the next selection.



Sampling without replacement

After selecting an element, we do not return it to the population before making the next selection.

Example:

A class has 30 students.

Select 5 students at random.

Once a student has been selected, that student cannot be selected again in the same sample.

Key difference

With replacement: the selected element can appear again.

Without replacement: the selected element cannot appear again.

Suppose we want to estimate the average amount of money City Tech students spend on transportation each month.

We survey 100 students.

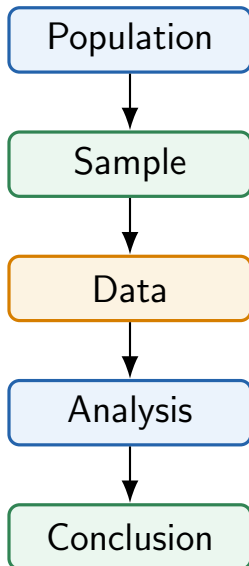
Identify:

- ① What is the population?
- ② What is the sample?
- ③ What is the variable?
- ④ Is the variable qualitative or quantitative?
- ⑤ If quantitative, is it discrete or continuous?

- **Population:** the City Tech students whose characteristic we want to study.
- **Sample:** the 100 students surveyed.
- **Variable:** monthly transportation spending.
- **Type:** quantitative.
- **Discrete or continuous:** generally treated as continuous when representing an amount of money.

Notice

We used concepts from both Sections 1.3 and 1.5 in the same problem.



Term	Meaning
Qualitative variable	A variable consisting of categories
Quantitative variable	A variable measured numerically
Discrete variable	A quantitative variable with countable values
Continuous variable	A quantitative variable that can take values over an interval
Population	All elements whose characteristics are being studied
Sample	A portion of the population selected for study

- ① First ask: **What kind of variable is this?**
- ② Qualitative variables describe **categories**.
- ③ Quantitative variables represent **numerical measurements**.
- ④ Discrete variables are generally associated with **counting**.
- ⑤ Continuous variables are generally associated with **measurement**.
- ⑥ The **population** is the full group we want to study.
- ⑦ The **sample** is the portion we actually study.
- ⑧ A good sample should be reasonably **representative** of the population.

Suppose we want to study the eating habits of City Tech students.

We survey 250 students and ask:

“How many meals did you eat yesterday?”

Identify:

- ① The population
- ② The sample
- ③ The variable
- ④ Qualitative or quantitative?
- ⑤ Discrete or continuous?

Suppose we want to study the eating habits of City Tech students.
We survey 250 students and ask:

“How many meals did you eat yesterday?”

Identify:

- ① The population
- ② The sample
- ③ The variable
- ④ Qualitative or quantitative?
- ⑤ Discrete or continuous?

Think carefully

The most important question is not “What calculation do I do?”
It is:

What exactly are my data describing?

Who?

What?

How?

Who are we studying?

Population / Sample

What are we measuring?

Variable

How does the variable behave?

Qualitative / Quantitative

If quantitative, how?

Discrete / Continuous

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Sections covered

Section 1.3 — Types of Variables

Section 1.5 — Population Versus Sample